

Familial Episodic Pain Syndrome With Predominantly Lower Limb Involvement — Comprehensive Disease Characterization

Autonomous discovery report — evidence base: human clinical case series/pedigrees, in vitro electrophysiology, and knock-in/knockout mouse models. Primary literature cited by PMID.

Summary (Answer to the Research Question)

"Familial Episodic Pain Syndrome With Predominantly Lower Limb Involvement" is the **SCN11A-related subtype** of familial episodic pain syndrome (FEPS3; OMIM #615552), an **autosomal-dominant sodium channelopathy of peripheral nociceptors**. It is caused by heterozygous **gain-of-function missense variants in SCN11A**, which encodes the voltage-gated sodium channel **NaV1.9** that sets the resting membrane potential (RMP) of small-diameter dorsal-root-ganglion (DRG) nociceptors. Gain-of-function shifts NaV1.9 activation to more hyperpolarized voltages, depolarizes the RMP, and renders nociceptors hyperexcitable, producing **early-childhood-onset, paroxysmal, cold-/fatigue-/weather-triggered episodic pain predominantly in the distal lower limbs** that characteristically **attenuates with age**. A closely related, phenotypically overlapping subtype (**FEPS2**, OMIM #615551) is caused by gain-of-function **SCN10A/NaV1.8** variants; FEPS1 (**TRPA1**, #615040) predominantly affects the upper body.

1. Disease Information

Overview. Familial episodic pain syndrome (FEPS) is a group of rare autosomal-dominant disorders characterized by **early-childhood onset of severe, self-limited episodes of pain affecting mainly the distal extremities, which tend to attenuate or diminish with age**

(Shen

2022,

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"Familial episodic pain syndromes (FEPS) are one of the distinctive syndromes characterized by early-childhood onset of severe episodic pain mainly affecting the distal extremities and tend to attenuate or diminish with age"). The subtype with **predominantly lower-limb involvement** corresponds to the tetrodotoxin-resistant sodium-channel FEPS subtypes — chiefly **FEPS3 (SCN11A/Nav1.9)**, with **FEPS2 (SCN10A/Nav1.8)** as a closely related alternative. In Japanese kindreds the syndrome was originally described as "infantile limb pain episodes" / paroxysmal limb intolerable pain, with pain most frequently in the **knees and ankles** (Noguchi 2017,).

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Key identifiers. - **OMIM:** #615552 — Episodic Pain Syndrome, Familial, 3 (FEPS3; *SCN11A*). Related: #615551 (FEPS2; *SCN10A*); #615040 (FEPS1; *TRPA1*). - **Gene OMIM:** *SCN11A* 604385; *SCN10A* 604427. - **MONDO (suggested):** MONDO term for "familial episodic pain syndrome 3" (*SCN11A*-related FEPS); parent "familial episodic pain syndrome." - **ICD-10:** best fit G90.09 / R52 (pain, disorders of autonomic/peripheral nervous system) — no dedicated code. **ICD-11:** 8D89 / MG30 (chronic/neuropathic pain categories) — no dedicated code. - **MeSH:** no dedicated descriptor; indexed under "Pain," "Sodium Channelopathies," "NAV1.9 Voltage-Gated Sodium Channel." - **Orphanet:** rare monogenic pain syndrome (ultra-rare; no established ORPHA prevalence class).

Synonyms / alternative names. Familial Episodic Pain Syndrome 3; FEPS3; *SCN11A*-related familial episodic pain; Nav1.9 channelopathy (episodic pain form); "infantile limb pain episodes"; paroxysmal limb intolerable pain; familial episodic limb pain. (FEPS2 synonyms: Familial Episodic Pain Syndrome 2; *SCN10A*/Nav1.8-related FEPS.)

Source of information. Derived from **aggregated disease-level resources and individual pedigree/case-series reports** (multigenerational families, mostly Japanese and Chinese, plus European cases), combined with in-vitro and animal functional studies — **not** from population EHR datasets.

2. Etiology

Disease causal factors — genetic. FEPS3 is a **monogenic channelopathy**. The primary cause is a heterozygous **gain-of-function (GoF) missense variant in *SCN11A*** (NaV1.9). FEPS2 is caused analogously by GoF *SCN10A* (NaV1.8) variants. There is **no infectious or neoplastic etiology**; environment acts only as a trigger of episodes, not as a cause.

Genetic risk factors (causal variants). Recurrent/known pathogenic *SCN11A* variants include **p.Arg222His (R222H)**, **p.Arg222Ser (R222S)**, **p.Arg225Cys (R225C)**, **p.Phe814Cys (F814C)**, **p.Phe1146Ser (F1146S)**, **p.Val1184Ala (V1184A)**, **p.Leu811Phe (L811F)**, and **p.Asn816Lys (N816K)** (Kabata 2018,

[P 30557356](#)

Huang 2019,

[P 31551682](#)

Nagao 2025,

[P 39058404](#)

Leipold 2015,

[P 26645915](#)

All are germline, heterozygous, autosomal-dominant.

Environmental risk factors. Not causal; **cold exposure, physical fatigue, and weather change** precipitate attacks (Noguchi 2017,

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Age is an inverse "risk factor" — episodes are most frequent in early childhood and diminish through adolescence.

Protective factors. No genetic protective alleles are established. **Environmental/behavioral protection:** warming the affected limb relieves pain, and avoidance of cold/fatigue reduces episode frequency (Noguchi 2017,

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In animal models, **transcutaneous cholesterol** reversed NaV1.9-mediated hypersensitivity

(Amsalem 2018,),
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 suggesting a candidate protective intervention.

Gene–environment interactions. The germline GoF variant lowers nociceptor threshold; **environmental modulators of NaV1.9 push already-sensitized neurons over threshold:** (i) cold — GoF confers **cold-resistant hyperexcitability** (Leipold 2015,);

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 (ii) inflammation lowers membrane cholesterol, moving NaV1.9 out of lipid rafts and enhancing its activation (Amsalem 2018,);

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 (iii) inflammatory mediators (bradykinin, ATP, PGE2, histamine, 5-HT) directly excite NaV1.9⁺ afferents (Hockley 2014,).

P 24972070

3. Phenotypes

Core phenotype (symptom). Paroxysmal, severe **episodic pain of the distal limbs, predominantly lower limbs** (knees, ankles > wrists, elbows). Episodes last **~15–30 minutes**, recur **several times per day**, are **triggered by fatigue and precede bad weather**, the affected region **feels cold**, and **warming relieves** the pain (Noguchi 2017,);

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"The pain typically lasts for 15-30 min and recurs several times a day. The pain is often induced by fatigue and is a prelude of bad weather. The affected regions feel cold in the patients, and warming the lesions relieves the symptoms.". - **HPO:** HP:0012531 (Pain); HP:0012532 (Chronic pain); HP:0025277 (Fluctuating course/episodic — episodic pain); HP:0003394 (limb pain — muscle pain proxy); HP:0002829 (Arthralgia — joint pain); HP:0001943 sensitivity.

Associated phenotypes (variant-dependent). - **Gastrointestinal:** episodic abdominal pain and chronic constipation with some variants (e.g., N816K), reflecting NaV1.9 expression in myenteric neurons (Huang 2019,

).

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HPO: HP:0002027 (Abdominal pain); HP:0002019 (Constipation). - **Cold intolerance / cold-aggravated pain** (Leipold 2015,

).

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HPO: HP:0500152 (Cold-induced pain/temperature sensitivity). - **Essential tremor co-segregation with R225C** (Leng 2017,

).

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HPO: HP:0001337 (Tremor).

Phenotype characteristics. - **Age of onset:** neonatal–early childhood/infancy (HP:0011463 childhood onset; often infantile onset). - **Severity:** moderate–severe pain during episodes; **variable** expressivity between and within families. - **Progression:** **episodic/fluctuating**, characteristically **spontaneously mitigating with age** (adolescence). - **Frequency among affected:** distal-limb pain is the defining feature (essentially 100% of affected); GI and tremor features are variant-specific minorities.

Quality-of-life impact. During childhood, recurrent daily painful episodes can disrupt sleep, activity, and schooling; because episodes are self-limited and remit with age, **long-term disability is low** and life expectancy is unaffected. No disease-specific QoL instrument (EQ-5D/SF-36/PROMIS) data are published.

4. Genetic / Molecular Information

Causal genes. - **SCN11A** — HGNC:10583; NCBI Gene 11280; Ensembl ENSG00000168356; UniProt **Q9UI33** (NaV1.9/SCN11A); chromosome **3p22.2**; OMIM gene 604385. Encodes the α -subunit of the tetrodotoxin-resistant voltage-gated sodium channel **NaV1.9**. - **SCN10A** (FEPS2) — HGNC:10582; NCBI Gene 6336; UniProt Q9Y5Y9 (NaV1.8); 3p22.2; OMIM 604427.

Pathogenic variants (SCN11A). Missense, germline, heterozygous. Representative: **R222H**, **R222S**, **R225C**, **F814C**, **F1146S**, **V1184A**, **L811F**, **N816K**. Located in functionally critical regions (e.g., L811 in the DII/S6 helix important for voltage-sensor→pore coupling; Nagao 2025,

).

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Classification: most reported variants are **Pathogenic/Likely Pathogenic** per ACMG

(segregation + functional GoF evidence); novel variants may be VUS pending functional testing. **Variant type/class:** point missense (no frameshift/nonsense/structural variants reported for the episodic-pain phenotype). **Allele frequency:** absent or ultra-rare in gnomAD (consistent with high penetrance and dominant pathogenicity). **Origin:** **germline** (no somatic/COSMIC relevance). **Functional consequence:** **gain of function** (hyperpolarized activation, larger persistent/window current, RMP depolarization).

Genotype–phenotype / allelic spectrum. *SCN11A* is pleiotropic: modest GoF → **FEPS3 / painful small-fiber neuropathy**; extreme GoF (sustained depolarization block) → **congenital insensitivity to pain (CIP)**. The same residue diverges by substitution: **L811Phe → FEPS, L811Pro** → **CIP** (Nagao 2025,).

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R225C additionally associates with **essential tremor** (Leng 2017,).

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Modifier genes. **PRMT7** methylates NaV1.9 (Arg519) and up-regulates NaV1.9 current and neuronal excitability — a candidate excitability modifier and drug target (Ma 2022,).

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Epigenetic information. No disease-specific DNA-methylation/histone changes reported. (Post-translational **arginine methylation** of the channel by PRMT7 is a molecular—not genomic—regulatory layer.)

Chromosomal abnormalities. None; FEPS3 is a single-nucleotide missense disorder, not a copy-number/structural condition.

5. Environmental Information

- **Environmental factors:** **Cold ambient temperature** and **falling barometric pressure / bad weather** trigger episodes (Noguchi 2017,).

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No toxin, radiation, or occupational exposure is implicated as a cause.

- **Lifestyle factors:** **Physical fatigue/exertion** is a major trigger; rest and warming mitigate. No association with smoking/alcohol/diet as causes.

- **Infectious agents: None** — not an infectious or immune-mediated disease.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **heterozygous germline gain-of-function missense variant in SCN11A** alters the NaV1.9 channel protein (*demonstrated: genetics + in vitro*).
2. The variant **shifts NaV1.9 voltage-dependence of activation to more hyperpolarized potentials and enlarges the persistent/window current** (*demonstrated: voltage-clamp; Leipold 2015; Huang 2019*) → **results in more channels open near rest**.
3. Increased subthreshold Na⁺ influx **depolarizes the resting membrane potential of small-diameter C-fiber DRG nociceptors** (*demonstrated: current-clamp in DRG neurons*).
4. RMP depolarization brings the membrane closer to firing threshold → **leads to a reduced action-potential threshold and increased firing frequency (nociceptor hyperexcitability)** (*demonstrated*); the hyperexcitability is **retained at cold temperatures** ("cold-resistant") (*demonstrated: Leipold 2015*).
5. Environmental modulators (cold, inflammation-driven membrane-cholesterol loss, inflammatory mediators, PRMT7 methylation) **further potentiate NaV1.9**, pushing sensitized neurons over threshold (*demonstrated in models; inferred as the trigger mechanism in patients*).
6. Hyperexcitable distal-limb nociceptive afferents generate **ectopic/amplified nociceptive signaling** (*inferred from cellular data*) →
7. **leads to the clinical episodic, cold-/fatigue-triggered distal (predominantly lower-limb) pain** (*clinical*).
8. **Branch (variant-dependent):** NaV1.9 in **myenteric/enteric neurons** → episodic abdominal pain + constipation (Huang 2019). NaV1.9 effects on broader neuronal excitability → **essential tremor** with R225C (Leng 2017).
9. **Branch (allelic):** extreme GoF → sustained depolarization/inactivation block → **loss of firing → congenital insensitivity to pain** (opposite phenotype).
10. **Age-dependent attenuation** (*clinical; mechanism inferred*) — episodes diminish through adolescence, possibly via developmental changes in channel expression/nociceptor maturation.

Detail by category

- **Molecular pathways / biochemical abnormality:** primary defect is an **ion-channel (voltage-gated Na⁺ channel) dysfunction**; NaV1.9 is the threshold channel with the most hyperpolarized activation among VGSCs and sets RMP (Köster 2025,

P 39836077

Downstream: action-potential electrogenesis and neurotransmitter release from nociceptor terminals. Modulation by inflammatory mediator signaling (bradykinin/ATP/PGE2), lipid-raft/cholesterol biology, and PRMT7 arginine-methylation.

- **Cellular processes:** neuronal membrane depolarization, action-potential generation, nociceptor sensitization/hyperexcitability (not apoptosis/inflammation-driven tissue destruction).
- **Protein dysfunction: gain of function** via altered gating (not misfolding/aggregation); molecular-dynamics work suggests some variants (L811F) increase structural stability and impede necessary conformational changes (Nagao 2025,

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- **Immune involvement:** none primary; inflammation is a **modulatory trigger**, not an autoimmune mechanism.
- **Tissue damage: none** — pain is due to aberrant neuronal signaling, not tissue injury; no fibrosis/necrosis/ischemia.
- **Suggested ontology terms:** GO:0086010 (membrane depolarization during action potential); GO:0019228 (neuronal action potential); GO:0035725 (sodium ion transmembrane transport); GO:0005248 (voltage-gated sodium channel activity, MF); GO:0050966 (detection of noxious stimulus). Cell types: **CL:0000198 (nociceptor)**, CL:0002249 / CL:0000101 (sensory/DRG neuron), CL:0011103 (enteric neuron). Subcellular: GO:0005886 (plasma membrane), GO:0045211 (axon/terminal), GO:0045121 (membrane raft).

7. Anatomical Structures Affected

- **Organ/system level: peripheral nervous system** (somatosensory), specifically primary sensory neurons; secondary involvement of the **enteric nervous system** (GI symptoms with some variants). Body system: **nervous (sensory)** and **digestive** (variant-dependent).

- **Tissue/cell level: nervous tissue** — small-diameter **C-fiber nociceptors** and **A δ** nociceptors of the **dorsal root ganglia** and **trigeminal ganglia**; **myenteric neurons** for GI features (Huang 2019,

P 31551682

 CL:0000198 (nociceptor); CL:0000101 (sensory neuron).
- **Subcellular level: neuronal plasma membrane**, including **cholesterol-rich lipid rafts** from which NaV1.9 redistributes upon sensitization (Amsalem 2018,

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 GO CC: GO:0005886, GO:0045121 (membrane raft).
- **Localization (UBERON): UBERON:0000044 (dorsal root ganglion)**; UBERON:0001132 (trigeminal ganglion); UBERON:0002240 (spinal cord dorsal horn, downstream); distal limb targets — **UBERON:0002103 (hindlimb)/UBERON:0003823 (leg)**, knees/ankles.
Lateralization: bilateral (both legs) and typically symmetric/migratory.

8. Temporal Development

- **Onset: congenital/infantile to early childhood**; onset pattern is **recurrent-paroxysmal (acute episodes on a chronic predisposition)**.
- **Progression: episodic/fluctuating, not progressive**; individual episodes are self-limited (~15–30 min). Over years the disorder **spontaneously attenuates**, typically remitting or markedly improving **around adolescence** (Noguchi 2017,

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 Shen 2022,

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).
- **Disease duration:** symptomatic phase is **childhood-limited** in most; the underlying genotype is lifelong and can manifest as milder adult sensitivity.
- **Remission:** predominantly **spontaneous/age-dependent**; treatment-induced relief is symptomatic.
- **Critical periods:** childhood (peak symptom burden and the window for supportive management and trigger avoidance).

9. Inheritance and Population

- **Inheritance: Autosomal dominant** (heterozygous). 50% offspring recurrence risk.
- **Penetrance / expressivity:** high but **variable penetrance; variable expressivity** in age of remission, episode frequency, and presence of GI/tremor features. No confirmed anticipation, mosaicism, or consanguinity requirement (dominant, not recessive).
- **Founder effect: p.R222H is a founder mutation in the Tohoku region of Japan** (7 of 48 families) (Kabata 2018,

P 30557356

"A founder mutation, SCN11A p.R222H was confirmed to be frequently observed in patients with FEP in the Tohoku region of Japan").

- **Carrier frequency:** not established; pathogenic alleles are ultra-rare/absent in gnomAD.
- **Epidemiology: ultra-rare;** no reliable prevalence/incidence per 100,000 is published. Reported cohorts are largely **Japanese** (≥ 48 families) and **Chinese**, with additional **European** cases.
- **Geographic distribution of variants:** R222H enriched in Tohoku, Japan; R225C reported in Chinese kindreds; V1184A (cold-aggravated) in European cases.
- **Sex ratio / age distribution:** autosomal dominant → **no strong sex bias ($\approx 1:1$)**; affected individuals are predominantly **children/adolescents**.

10. Diagnostics

- **Clinical diagnosis** rests on the **characteristic history:** autosomal-dominant family history + early-childhood recurrent, self-limited, cold/fatigue-triggered distal (lower-limb) pain relieved by warmth. No specific routine **laboratory** abnormality or **biomarker**; inflammatory markers, imaging (X-ray/MRI), and standard **nerve conduction studies** are typically **normal** (large myelinated fibers spared) — useful mainly to exclude alternatives.
- **Electrophysiology:** conventional NCS usually normal; research-level microneurography/skin biopsy for small-fiber assessment may be considered (small-fiber pathology overlaps the SCN11A spectrum).

- **Genetic testing (definitive): single-gene sequencing or a hereditary-pain/sodium-channelopathy gene panel** including *SCN11A*, *SCN10A*, *SCN9A*, **TRPA1*; *WES/WGS* for atypical/panel-negative cases. *CMA*, *karyotyping*, *FISH*, *mtDNA*, and *repeat-expansion testing* are not indicated (*point-mutation disorder*). *Variant interpretation benefits from functional (electrophysiology) confirmation** given the GoF/LoF-to-phenotype divergence.
- **Diagnostic criteria:** no formal society criteria; diagnosis is molecular + clinical.
- **Differential diagnosis: growing pains** (benign, no family-segregating molecular cause), **erythromelalgia / inherited erythromelalgia (SCN9A)** — but erythromelalgia features red, hot, hyperperfused extremities (opposite of the cold sensation here) (Klein-Weigel 2018, [P 29299961](#)), **paroxysmal extreme pain disorder (SCN9A)**, **Fabry disease**, **small-fiber neuropathy**, juvenile idiopathic arthritis, and autoinflammatory periodic syndromes.
- **Screening: cascade genetic testing** of at-risk relatives once a familial variant is identified. No newborn/population screening.

11. Outcome / Prognosis

- **Survival/mortality: normal life expectancy; no disease-specific mortality.** Not a life-threatening condition.
 - **Morbidity/function:** morbidity is **episodic childhood pain** with possible sleep/activity disruption; **long-term disability is minimal** because episodes remit with age.
 - **Disease course/complications:** benign natural history with **spontaneous mitigation around adolescence**; complications limited to variant-specific GI symptoms (constipation) or comorbid essential tremor (R225C).
 - **Recovery potential:** high — most patients improve substantially without disease-modifying therapy.
 - **Prognostic factors:** specific variant (biophysical severity), presence of GI/tremor features; **age** is the dominant prognostic factor (improvement over time). No validated prognostic biomarker.
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12. Treatment

No disease-specific approved therapy; management is symptomatic and the condition is largely self-limiting.

- **Supportive care (first-line):** warming affected limbs (relieves pain), **avoidance of cold and fatigue** (Noguchi 2017,

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NCIT: Supportive Care (C15272).

- **Pharmacotherapy (off-label, symptomatic):** simple analgesics/NSAIDs; **state-dependent sodium-channel blockers** — **carbamazepine, lidocaine, mexiletine**. Carbamazepine can act as an **activation modulator** that partially normalizes gain-of-function sodium-channel gating (Han 2018,

P 30135145

NCIT: Carbamazepine (C376), Lidocaine, Mexiletine, NSAID (C198).

- **Pharmacogenomics / precision medicine:** genotype-guided rationale — GoF NaV1.9 is the direct target; **NaV1.9-selective blockers are investigational** (preclinical/early development), highlighted for cold pain and small-fiber neuropathy (Chen 2025,

P 40206072

PRMT7 inhibition (e.g., DS-437) reduces NaV1.9 currents and pain hypersensitivity in mice — a candidate strategy (Ma 2022,

P 34326297

Transcutaneous **cholesterol** reversed NaV1.9-mediated hypersensitivity in models (Amsalem 2018,

P 29459435

- **Advanced therapeutics (gene/cell/RNA):** none in clinical use; conceptually amenable to allele-selective ASO/gene-editing given the dominant GoF mechanism (future direction).
- **Surgical/interventional:** not applicable.
- **Treatment outcomes/adverse events:** no controlled trial data; management guided by general neuropathic-pain practice and drug-specific safety profiles.

13. Prevention

- **Primary prevention:** not possible (germline genetic cause); **preconception/prenatal genetic counseling** and options (PGT/prenatal testing) can prevent transmission where desired.
 - **Secondary prevention: cascade genetic testing** of relatives for early recognition; early trigger-avoidance counseling.
 - **Tertiary prevention:** minimize episode burden via **avoidance of cold/fatigue** and prompt warming; manage variant-specific complications (e.g., constipation).
 - **Immunization / public-health / environmental interventions:** not applicable.
 - **Counseling: genetic counseling** for autosomal-dominant 50% recurrence risk and variable expressivity is central. NCIT: Genetic Counseling (C15311).
 - **Prophylaxis:** no established prophylactic drug; behavioral trigger avoidance is the mainstay.
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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** human *SCN11A* (NaV1.9). Mouse ***Scn11a*** — NCBI Gene 24046; MGI:1345149 (Chr 9). Rat *Scn11a* ortholog exists. NaV1.9 (originally NaN/SNS2) is **evolutionarily conserved** across mammals with conserved nociceptor-restricted expression.
 - **Natural disease in other species: no naturally occurring animal disease** cataloged in OMIA for SCN11A episodic pain; disease knowledge in animals comes from **engineered models**, not spontaneous veterinary cases.
 - **Comparative biology:** mouse NaV1.9 gating and DRG expression closely parallel human, enabling faithful modeling; **conservation of the RMP-setting/threshold role** underlies the translational validity.
 - **Transmission:** not applicable (non-infectious, non-zoonotic).
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15. Model Organisms

- **Model type:** mammalian — **mouse (*Mus musculus*, NCBI Taxon 10090)**; plus in-vitro heterologous expression (HEK293, DRG electroporation) and patient-derived DRG-neuron electrophysiology.
- **Genetic models:**
- **Knock-in (disease-variant) mice: *Scn11a*-R222S** — hypersensitive to hot and cold stimuli, increased DRG input impedance and evoked AP firing (Noguchi 2017,

P 30549873

F802C and **F1125S** (orthologues of human F814C/F1146S) — elevated RMP and increased AP firing to high input current; F1125S also increases firing probability at low input (Kabata 2018,

P 30557356

A796G knock-in used in modifier studies (Ma 2022,

P 34326297
- **Knockout (loss-of-function) mice: *Nav1.9*-null** — reduced nociceptor excitability, +55% C-fiber electrical threshold, higher mechanical thresholds, impaired noxious heat sensing (Hoffmann 2017,

P 27780178

attenuated visceral afferent activation (Hockley 2014,

P 24972070

Nav1.8/Nav1.9 double-KO mildly affects acute pain and deregulates C-LTMR/neuropeptide genes (Alves-Simões 2025,

P 39382328
- **Phenotype recapitulation: strong** — GoF knock-ins reproduce the cardinal DRG hyperexcitability and thermal/cold hypersensitivity; reciprocal KO produces the opposite (hyposensitivity), establishing causality bidirectionally.
- **Limitations:** mice model **peripheral electrophysiology and behavioral proxies**, not the subjective, episodic, weather-triggered nature of human pain or its spontaneous adolescent remission; genetic background and orthologue-residue differences require care.

- **Applications:** target validation, variant functional classification, and testing of NaV1.9-directed and modifier (PRMT7, cholesterol) therapeutics.
- **Resources:** MGI/IMPC/IMSR for *Scn11a* alleles.

Supported and Refuted Hypotheses

Supported: 1. The lower-limb-predominant FEPS is a **NaV1.9 (SCN11A) gain-of-function channelopathy** (multiple kindreds + electrophysiology + mouse models). 2. **Nociceptor RMP depolarization → hyperexcitability** is the mechanistic core (in vitro + KI/KO). 3. **Autosomal-dominant** inheritance with a **Japanese R222H founder** effect. 4. **Cold/fatigue trigger episodes via physiological modulation of NaV1.9**. 5. **Benign, age-attenuating natural history**.

Refuted / excluded: - Not caused by tissue damage, inflammation-as-primary, infection, or structural/chromosomal abnormality. - Not an erythromelalgia phenotype (patients feel **cold**, relieved by warmth — the inverse of erythromelalgia).

Limitations & Future Directions

- No population-level prevalence, sex-ratio, or QoL-instrument data; cohorts are small and geographically skewed (Japan/China).
- Ambiguity in whether the exact source label ("predominantly lower limb") intends **FEPS3 (SCN11A)** vs **FEPS2 (SCN10A)** — both cause distal, predominantly lower-limb episodic pain; FEPS3 is the best-characterized match and the primary focus here, with FEPS2 covered as the close alternative.
- No controlled treatment trials; **NaV1.9-selective analgesics and allele-selective genetic therapies** are the key future opportunities.

Evidence source key: human clinical (pedigrees/case series) — PMIDs 30549873, 30557356, 31551682, 28298626, 39058404, 26645915; in vitro/electrophysiology — 39836077, 30135145, 34326297, 29459435; mouse models — 30549873, 30557356, 27780178, 24972070, 39382328; reviews — 36051609, 29299961, 40206072.

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